An AI-Assisted Chest X-ray Training Companion for Student Radiologists

Sharable Diagrams Pack — Use Case • Activity • Sequence

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1 Use Case Diagram

Normative source: Figure 3.1 (Methodology) → Figure 4.1 context in Design. System boundary = Training Companion System.

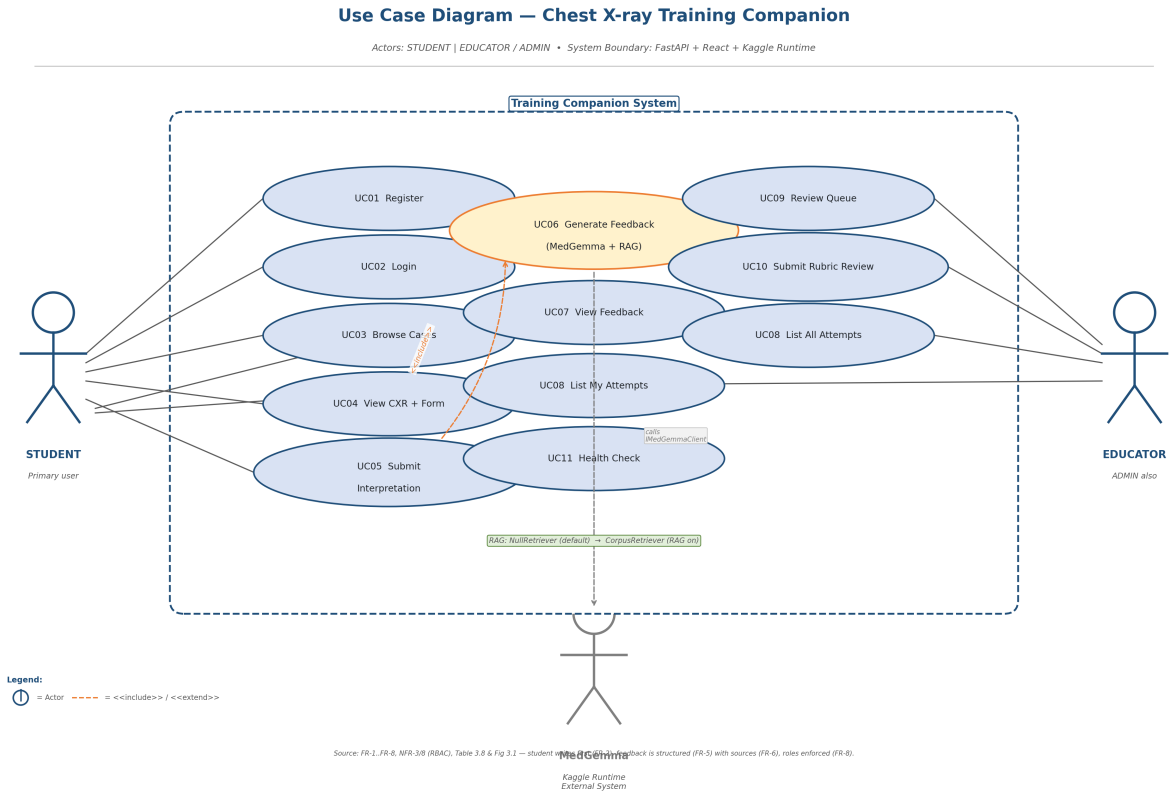


Figure 1 — Use Case Diagram (UC01–UC11). Amber dashed = UC05 <<include>> UC06; grey dashed = IMedGemmaClient → Kaggle Runtime.

Actors: STUDENT (primary) owns Register/Login/Browse/View/Submit/View Feedback; EDUCATOR/ADMIN owns Review Queue + Submit Rubric Review; MedGemma/Kaggle is an external system actor called only via IMedGemmaClient. All human use-cases pass through FastAPI routes (Table 3.11) and are orchestrated by ports (Table 3.10). RBAC is enforced by Role enum + require_role(...) (FR-8/NFR-3). RAG is a swappable IRetriever port (Null → Corpus, k=3).

2 Activity Diagram (Swimlanes)

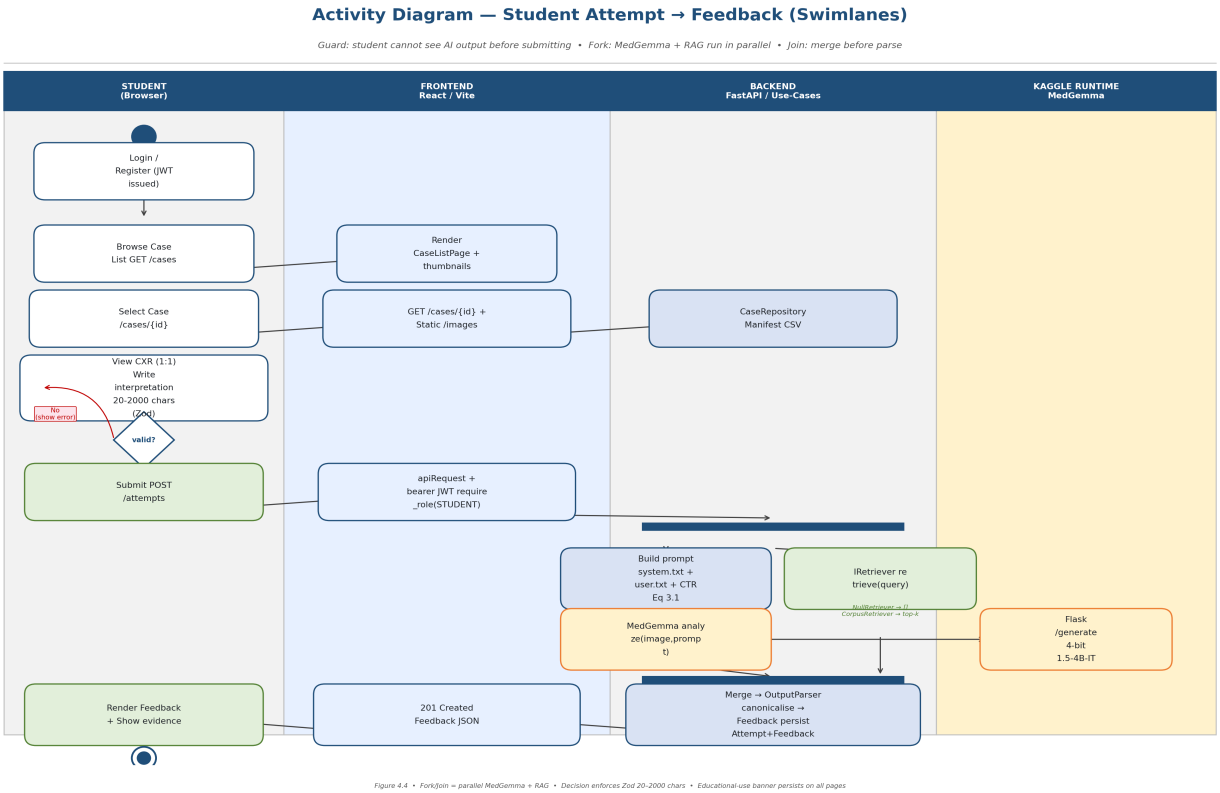


Figure 2 — Activity Diagram. Thick blue bars = fork/join for parallel MedGemma + RAG; red diamond = Zod 20–2000 char guard. The guard that feedback is hidden until Submit is both a UI invariant (submit is the only transition) and an API invariant (no pre-generated feedback is exposed).

Flow: Login → Browse CaseListPage (GET /cases) → Select Case (GET /cases/{id} + /images) → View CXR + write interpretation → valid? No → Zod error loop; Yes → POST /attempts (apiRequest + bearer JWT, single-flight 401 refresh) → Backend fork: (a) Build prompt (system.txt + user.txt + CTR Eq 3.1) → MedGemma → Flask /generate, (b) IRetriever (Null → [] else top-k). Join → Merge → OutputParser (strip ```json, balanced-brace JSON, FindingLabel alias map, 500 ch cap) → persist Attempt+Feedback (Feedback.raw carries promptVersion/modelVersion/passageIds) → 201 + Feedback JSON → Render FeedbackSection + Show evidence.

3 Sequence Diagram

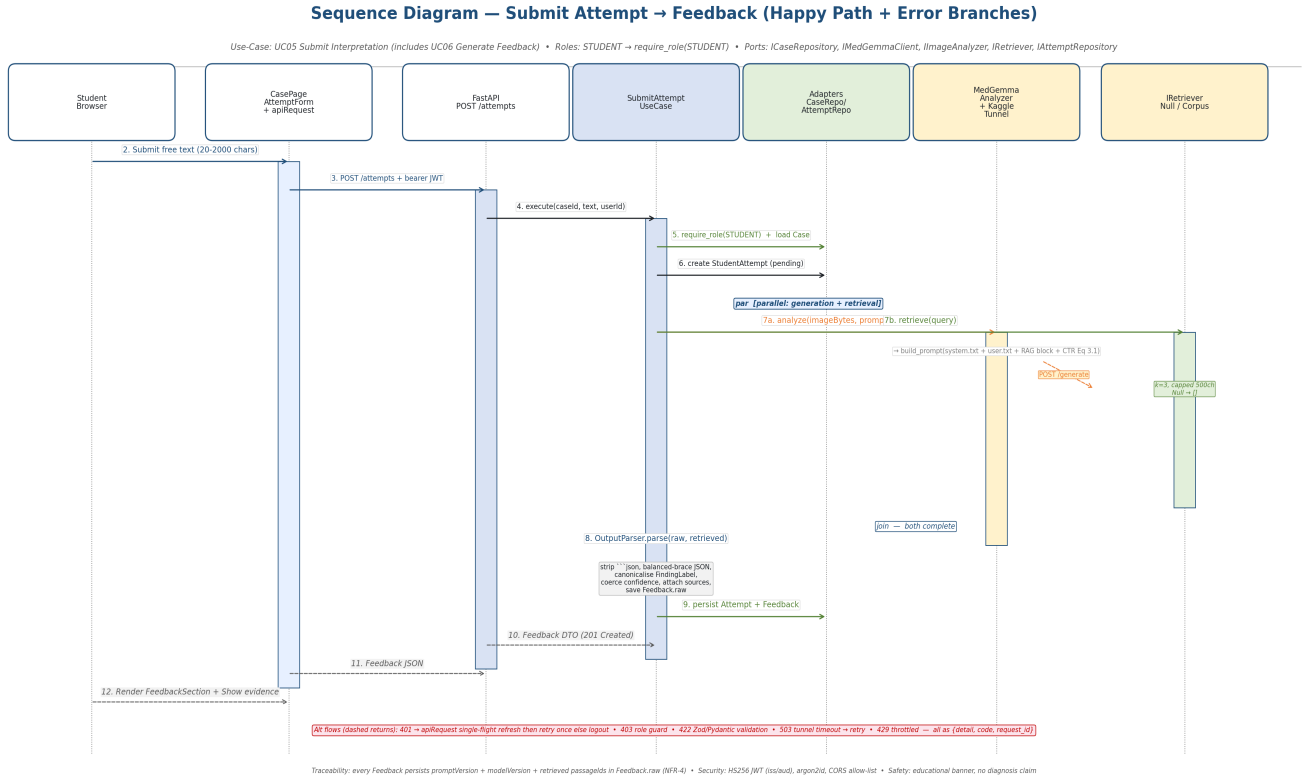


Figure 3 — Sequence Diagram for Submit Attempt (UC05 <> UC06). Solid = call, dashed = return, par = parallel generation + retrieval. Lifeline names equal module names for code navigation.

Happy path: Student → CasePage (Submit) → FastAPI POST /attempts → SubmitAttemptUseCase.execute → require_role(STUDENT) + load Case + create StudentAttempt → par: analyze(imageBytes, prompt) → Kaggle → POST /generate and retrieve(query) → join → OutputParser.parse → persist → 201 Feedback DTO → FeedbackSection. Alt flows: 401 (refresh once else logout), 403 (role guard), 422 (validation), 503 (tunnel timeout → retry), 429 (throttled) — all as {detail, code, request_id}. Traceability (NFR-4) is the Feedback.raw record; safety (NFR-2) is the parser boundary that never exposes raw model text.

Traceability & files: FR-1 (Display CXR) → UC03/04 → manifest_repository.py; FR-2 (write first) → UC04/05 guard → AttemptForm.tsx; FR-3 (MedGemma) → medgemma_analyzer.py; FR-4 (RAG) → IRetriever; FR-5/6 (feedback + sources) → feedback.py / FeedbackSection.tsx; FR-7 (review) → ReviewQueuePage; FR-8 (RBAC) → user.py / dependencies.py. Regeneration: python gen_diagrams.py → python build_diagrams_doc.py (matplotlib only, no Graphviz).